Dashboard for Historical Digital Business Data

Client: Club Data Department

Contractor: [Group Name] – Group E

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1. Context

In today's digital business environment, decision-making increasingly depends on the ability to analyze large volumes of information quickly, visually, and comprehensibly.

However, the club's digital analytics historical data with its key indicators—such as web traffic, conversion, online sales, or user behavior—is still maintained in traditional spreadsheets due to the large number of different sources from which the information is extracted.

Although Excel is a versatile tool, using it to manage complex historical series often results in files that are difficult to maintain, poorly scalable, and vulnerable to human error.

Given this situation, the digital analytics team needs to design and develop a visualization tool that allows them to organize, centralize, and effectively display the main digital business indicators.

The goal is to transform scattered and unwieldy data into interactive dashboards that facilitate the interpretation of trends, comparative analysis, and the rapid identification of insights and improvement opportunities. The tool should pursue not only visual clarity but also agility when displaying and sharing data with business areas, while reducing the analyst's time spent on data analysis.

The expected outcome is a robust, intuitive, and scalable system that lays the foundation for having an insight extraction and analysis workflow that is as automated as possible.

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2. Problem Statement

Our digital analysts need to quickly answer questions such as:

- What is the trend of the main metrics compared to previous seasons?
- Has any historical record been set in the last month?
- How does our digital performance compare to other European clubs?
- Are there significant drops or abrupt trend changes?
- Are we acquiring new users at the same pace as our competitors?
- Does the conversion rate indicate that our digital assets are optimized?

Currently, answering any of these questions requires the analyst to manually review the data and prepare ad-hoc reports to evaluate the results.

We need a solution that allows analysts to reduce their workload.

Data updates in Excel will remain manual, but we need a tool where we can work with that data effectively.

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3. Scope of Work

We are contracting your team to design, build, and deliver a Digital Analytics Visualization Tool under the following specifications:

3.1 Data

- You will receive in Excel format the complete historical data divided by digital asset:
    - **Main Website:** The club's corporate website; it is a content site aimed at keeping fans informed about the club's latest news. Its business objectives are user acquisition and engagement.
    - **eCommerce:** A portal for selling jerseys and club merchandise. Its business objective is sales.
    - **Streaming:** A multimedia content viewing platform (highlights, matches, live broadcasts, etc.), intended for video consumption and increasing fan engagement with the club.
    - **Fan App:** A mobile application designed for live match consumption. Its objective is the acquisition of new fans and match-day follow-up.
- The Excel file includes a description of each metric and its category.
- A separate Excel file is included with some metrics compared against competitor clubs.

3.2 Data Pipeline

Build a reproducible data pipeline in Databricks that:

1. Ingest the raw tracking data.
2. Calculates the metrics and fields needed to display the data in an optimized manner.
3. Generates a clean dataset (Delta table) suitable for subsequent visualization.

The pipeline must be modular, documented, and deployable—another engineer on our team should be able to run it with a different match with minimal changes.

3.3 Visualization

Deliver a Power BI dashboard connected to the Databricks output that displays, at a minimum:

Layer	Description
Summary Sheet	At a single glance, we should be able to see the status of the main metrics across the different businesses for the selected period: month, season, or year. Key metrics for each business should be selected, and in a visual and usable way, the analyst should have an overview of the business status. For season and year groupings, the monthly average of the selected time period will be analyzed.
Detail Sheets per Digital Asset	We must be able to analyze each digital asset separately with all the key indicators found in its Excel sheet. Metrics should be analyzable grouped by month, season (July through June), and calendar year. It will be important to visualize metrics by category: performance, attribution, or quality. Comparisons with prior periods and charts showing temporal evolution will be added. For season and year groupings, the monthly average of the selected time period will be analyzed. For those metrics where comparison with other clubs is available, comparative data must be included.
Metric Detail	On an individual sheet, we will have the ability to analyze a single metric in detail, being able to examine its temporal evolution (month, season, year), its variation compared to prior periods, and its comparison with competitor clubs (if available). It should also be possible to compare across digital assets.

The design, sheet/detail structure, filters required on each sheet, groupings, and the Power BI navigation structure will be defined by the students.

Students will need to choose the appropriate visualizations for each type of data to be displayed and give prominence to those that are most important for each type of business.

3.4 Bonus — Include AI for Automatic Insight Calculation

As added value, implement an AI integration (to be chosen by the students) that allows the analyst to automatically identify:

- Records or sharp declines in the last month or analysis period.
- A general summary of the status of the digital asset in the last month or analysis period.
- Quick insights based on historical data.

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4. Technical Environment

Component	Specification
Compute and Storage	Databricks workspace (free edition)
Language	Python (PySpark where appropriate)
Visualization	Power BI (mandatory); Databricks App (bonus)
Version Control	Git repository (hosted in the Databricks workspace or on GitHub)

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5. Deliverables

#	Deliverable	Format
1	Deployable pipeline code	Databricks notebooks and/or repositories, fully documented
2	Power BI report	.pbix file + instructions for connecting to the Databricks data source
3	Databricks App (bonus)	Deployable application package with source code
4	Technical documentation	Markdown or PDF covering architecture, data model, methodology, and user guide
5	Final presentation	Executive-level PowerPoint + live demo (15–20 min presentation + 10 min Q&A)

5.1 Tutoring Sessions with Experts

Each team will have access to a maximum of 4 hours of tutoring with an expert from the Club's Data Department throughout the project duration. Teams must propose a tutoring schedule as part of their Week 2 work plan.

We recommend distributing the sessions as follows (indicative, not mandatory):

Session	Suggested Timing	Suggested Duration	Focus
1	End of Week 2	45 min	Validate approach and data understanding

Session	Suggested Timing	Suggested Duration	Focus
2	End of Week 4	60 min	Pipeline review and model validation
3	Mid-Week 6	60 min	Visualization feedback and what-if design
4	Mid-Week 7	45 min	Pre-delivery review and presentation rehearsal

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6. Work Plan Submission (Week 2)

At the end of Week 2, each team must submit a document containing:

1. **Team Identity** — Team name and a brief description of the team's approach.
2. **Roles and Responsibilities** — The role assigned to each team member (e.g., Project Lead, Data Engineer, Visualization Specialist, ML/Analytics Lead) with a clear description of their responsibilities.
3. **8-Week Activity Plan** — A Gantt chart or equivalent breakdown of tasks by role and by week.
4. **Tutoring Schedule** — Proposed dates, durations, and agenda items for each of the (up to) 4 tutoring sessions.

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7. Licensing

All code and deliverables must be submitted under an open-source license, such as the Apache License, Version 2.0.

— *Club Data Department*